

Muhammad Asadullah

Principal Software Engineer

asad@asadcodes.com | +92 316 4363605 | linkedin.com/in/i-asad | github.com/iasad95 | asadcodes.com

SUMMARY

Principal Software Engineer with 8+ years building cloud-native SaaS and AI-enabled products using Node.js, Python, React, Next.js, Angular, and AWS. Architected systems serving 12M+ users, including a file platform processing 500K+ operations daily; reduced API latency by 45%; led a zero-downtime migration of 50M+ records; and designed a centralized AI middleware and MCP server layer for Claude/OpenAI-based document intelligence and conversational workflows.

CORE SKILLS

Backend: Node.js, NestJS, Express.js, Python, FastAPI, REST APIs, GraphQL, JWT/OAuth 2.0

Frontend: React, Next.js, Angular, Redux, NgRx, RxJS, TypeScript, Tailwind CSS

Databases: MongoDB, PostgreSQL, MySQL, Redis

Cloud & DevOps: AWS (Lambda, S3, EC2, RDS), Docker, Kubernetes, CI/CD

Architecture: Microservices, Distributed Systems, Event-Driven, CQRS, System Design

AI / LLM: Claude & OpenAI APIs, MCP Server Development, RAG & Vector Search, LangChain, n8n, Structured Outputs & Tool Calling

Testing: Jest, Cypress, Unit/Integration/E2E Testing, TDD

Tools: Git, Jira, Swagger, OpenAPI, PM2

Also comfortable with: C#, .NET, ASP.NET Core, Entity Framework, and Django.

EXPERIENCE

Principal Software Engineer, Biztree

Jan 2023 - Present | Remote, Canada

- Architected unified microservices backend for 12M+ users, cutting backend effort by 40%
- Designed AI middleware & MCP server layer: Claude/OpenAI workflows with secure tool access and token tracking
- Built CloudDrive: 500K+ daily file operations with RBAC and secure access
- Led zero-downtime MongoDB migration: 50M+ records with rollback-safe batch processing
- Reduced API latency by 45% using BullMQ and RabbitMQ
- Built HRM backend with PostgreSQL, TypeORM, CQRS
- Strengthened engineering foundations: Docker, CI/CD, structured logging, and automated testing

Senior Software Engineer, ConvertSite (ConvertCalculator), Part-time Contract

Jun 2024 - Aug 2025 | Remote, Netherlands

- Evolved a drag-and-drop builder into a prompt-driven AI app generator
- Generated calculators, quote forms, and landing pages directly from user intent
- Built a rules-driven workflow engine for pricing logic and lead capture
- Shipped embeddable integrations for WordPress, Shopify, Wix, and Framer
- Delivered full-stack features on a PostgreSQL-backed platform

Senior Software Engineer, InvoZone

Oct 2021 - Jan 2023 | Lahore, Pakistan

- Built enterprise PAM platform: credential and session management
- Designed scalable microservices with load balancing and clustering
- Added session forensics: keystroke and video recording for auditing
- Improved deployment speed by 30% with CI/CD automation
- Hardened security: XSS, SQL injection, brute-force protection
- Won Globee Gold Award for security innovation (2022)

Software Engineer, TenX

Aug 2018 - Oct 2021 | Lahore, Pakistan

- Built automated contamination detection system, reducing processing time by 90%
- Built real-time golf performance tracking with Microsoft, improving accuracy by 40%
- Designed distributed systems for high-throughput scientific computing
- Doubled release frequency with CI/CD automation and testing
- Optimized internal NPM packages to improve developer productivity
- Won #3 ranking in global sports innovation awards (Arccos)

EDUCATION & CERTIFICATION

Bachelor of Science in Computer Science (BSCS)

FAST-NUCES, Lahore, Pakistan · 2014 - 2018

AWS Certified Developer – Associate

Amazon Web Services